Indian Institute of Science Education and Research Kolkata

Department of Mathematics and Statistics

Subject: Probability and Statistics-I (MA2104)

Mid-semester Examination Full marks: 20

Date of Examination : 10.09.2025 Max. time: 90 minutes

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Instructions: There are four questions in all. Answer all the questions. In order to get full credit, all the intermediate steps must be provided and the answers must be given in its most simplified forms. Every answer must be explained with proper reasoning. The notations have their usual meanings. The number inside the square brackets at the right of each question denotes the marks corresponding to that question.

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1. The joint probability distribution of X and Y is given below.

$Y\downarrow$ $X\rightarrow$	-1	1
1	1/8	3/8
2	1/4	1/4

   (a) Find the marginal distributions of X and Y. What can you say about measure of skewness of the distribution of Y? [2+1=3]

   (b) Find the conditional distribution of $(Y|X=1)$. [1]

   (c) What is $E(X|Y=2)$? [1]

   (d) Find $Cov(X,Y)$. Are X and Y independent? [1+1=2]

2. When is a function $F:\mathbb{R}\rightarrow\mathbb{R}$ called the cumulative distribution function (cdf) of a random variable X? Verify whether the following F is a cdf. If yes, find the corresponding probability function. [$2+2+2=6$]

$$F(x)=\begin{cases} 0 & \text{if } x<0 \\ \frac{x^2}{4} & \text{if } 0\leqslant x<1 \\ \frac{1}{4} & \text{if } 1\leqslant x<2 \\ \frac{x^2}{3}-x+1 & \text{if } 2\leqslant x<3 \\ 1 & \text{if } x\geqslant 3. \end{cases}$$

3. Let F be the cdf of a continuous random variable X such that

$$P(X\leqslant x)+P(X\leqslant -x)=1 \quad \text{for all } x\in\mathbb{R}.$$

   (a) If it is known that the moments exist, find $E(X)$. [2]

   (b) What is the value of μ_3? [2]

4. Let X and Y be two independent random variables such that $X\sim bin\left(4,\frac{1}{2}\right)$ and $Y\sim bin\left(3,\frac{2}{3}\right)$. Determine $P(X-Y=3)$. Evaluate $E(XY)$. [2+1=3]

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